CH. BANSI LAL POLYTECHNIC, BHIWANI

DEPARTMENT OF COMPUTER ENGINEERING

(Affiliated to Haryana State Board of Technical Education, Panchkula)

A MAJOR PROJECT REPORT ON

“AXOLYT ORGANIC ENERGY”

A Modern Cloud-Native Botanical E-Commerce & 5-Stage Cold-Chain Tracking Platform

Submitted in partial fulfillment of the academic requirements
for the award of the Diploma of

DIPLOMA IN COMPUTER ENGINEERING

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CERTIFICATE OF APPROVAL

This is to certify that the project entitled "AXOLYT ORGANIC ENERGY — A Modern Cloud-Native Botanical E-Commerce & Supply-Chain Tracking Web Application" has been successfully executed and submitted by PRINCE KUMAR, bearing Roll No. 240790825012, in partial fulfillment of the requirements for the award of the Diploma in Computer Engineering from Ch. Bansi Lal Polytechnic, Bhiwani, during the academic session 2026–2027.

This report reflects original, authentic, and high-quality software engineering work carried out under my direct supervision and guidance. The project incorporates modern enterprise web architectures including React 19, TypeScript, Tailwind CSS v4, Motion Physics, and Google Firebase Cloud Firestore with custom role-based security rules and order gating.

To the best of my knowledge and professional evaluation, this project report or any substantial part thereof has not been submitted to any other university, board, or institution for the award of any diploma, degree, or academic credential.

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CANDIDATE'S DECLARATION

I, PRINCE KUMAR, student of Diploma in Computer Engineering at Ch. Bansi Lal Polytechnic, Bhiwani, hereby solemnly declare that the project entitled "AXOLYT ORGANIC ENERGY" submitted to the Department of Computer Engineering is a record of original, authentic, and independent development work carried out by me under the supervision and guidance of Ms. Pratibha Dhaiya, Lecturer in Computer Engineering.

I confirm that:

- **Authenticity of Implementation:** The entire source code, system architecture, database modeling, security rules, and user interfaces represent my authentic engineering effort and genuine system design.
- **No Plagiarism:** The practical implementation and written report contain no plagiarized material from any other student project or unauthorized source.
- **Real Integration:** All features detailed in this report, including Firebase Authentication, Firestore NoSQL database operations, Cold-chain supply tracking, and the Admin Command Center, are functional and operational.
- **Compliance:** I have strictly adhered to the academic, ethical, and professional guidelines established by Ch. Bansi Lal Polytechnic and the State Board of Technical Education.

Date: 08 November 2026
Place: Bhiwani, Haryana

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ABSTRACT

In contemporary digital commerce, consumers increasingly demand full ingredient provenance, sustainable packaging, and transparent supply chains, particularly within the organic food and functional beverage industries. Traditional e-commerce platforms frequently fail to bridge the gap between premium brand storytelling, real-time logistics transparency, and strict data security.

This project presents AXOLYT ORGANIC ENERGY, an enterprise-grade, full-stack Single Page Application (SPA) designed to deliver an elevated e-commerce experience for cold-pressed, adaptogenic, and plant-based energy beverages. Developed using React 19, TypeScript, Tailwind CSS v4, and Google Firebase Cloud Firestore, the platform unites modern sensory web design with rigorous cloud data security.

Key technical achievements of this project include:

• **Strict Authentication Gating:** Enforces zero guest checkouts by automatically preserving uncompleted shopping sessions and launching the authentication portal, preventing orphaned transactions.

• **Real-Time Supply Chain Tracking:** Features an interactive 5-stage cold-chain delivery tracker (Confirmed -> Chilled Packaging -> Clean Line-Haul -> Local Route -> Doorstep Verification) with live step progression.

• **Full-Featured Admin Command Center:** Equips store administrators with real-time analytics, order lifecycle management with deletion permissions, multi-flavor inventory monitoring, and live store banner announcements.

• **3D Kinetic Botanical Physics:** Implements custom interactive tilt-shift cards utilizing Motion 12 spring mechanics, inverted parallax textures, and specular light sheen.

• **Cloud Firestore RBAC Security:** Enforces strict collection-level rules preventing cross-user data leaks and restricting administrative mutations to authenticated master accounts.

The resultant platform successfully demonstrates how modern web engineering principles can be synthesized to produce a secure, high-performance, and visually distinctive enterprise web application.

TABLE OF CONTENTS

Certificate of Approval 2
Candidate's Declaration 3
Acknowledgement 4
Abstract 5
List of Figures, Tables & Code Listings 7
Chapter 1: Introduction & Project Overview **8**
Chapter 2: Problem Statement, Objectives & Feasibility Study **9**
Chapter 3: Scope of the Project & Target User Personas **10**
Chapter 4: Hardware & Software Requirements (Tables 4.1 & 4.2) **11**
Chapter 5: Technologies Used — Frontend (React 19, TS, Tailwind, Motion) **12**
Chapter 5 (Cont.): Cloud Infrastructure & Web Audio Ambiance Synthesis **13**
Chapter 6: System Analysis & Software Requirements Specification (SRS) **14**
Chapter 7: System Architecture & 3-Tier Component Design **15**
Chapter 8: Database Design & Firestore NoSQL Schema (Tables 8.1 & 8.2) **16**
Chapter 8 (Cont.): Cloud Security Rules & Role-Based Access Control (Code 1) **17**
Chapter 9: Data Flow Diagrams (Level 0 Context & Level 1 DFD) **18**
Chapter 9 (Cont.): Use Case Specification Matrix & Gated Checkout Flow **19**
Chapter 10: Core Modules: Navigation & Web Audio Engine (Code 2) **20**
Chapter 10 (Cont.): Core Modules: 3D Tilt Cards & Gated Checkout Guard (Code 3 & 4) **21**
Chapter 11: Website Output Screenshots: Hero & Ingredients (Figures 11.1 & 11.2) **22**
Chapter 11 (Cont.): Website Output Screenshots: Admin Portal & Mobile (Figures 11.3–11.5) **23**
Chapter 12: Core Modules: 5-Stage Cold-Chain Tracker & Admin Hub (Code 5 & 6) **24**
Chapter 13: Software Testing & Comprehensive Test Cases Matrix (Table 13.1) **25**
Chapter 14: Platform Advantages & Technical Limitations **26**
Chapter 14 (Cont.): Future Scope & Planned Engineering Enhancements **27**
Chapter 15: Conclusion & Academic Learnings **28**
References & Bibliography 29

FIGURES, TABLES & CODE LISTINGS

LIST OF FIGURES & OUTPUT SCREENSHOTS

Figure 7.1: Three-Tier System Architecture Model (Presentation, Logic, Cloud) 15
Figure 9.1: Data Flow Diagram Level 0 (Context Interaction Model) 18
Figure 9.2: Data Flow Diagram Level 1 (Transactional Process Breakdown) 18
Figure 11.1: Website Output: Homepage Hero Banner & Forest Waterfall Ambiance 22
Figure 11.2: Website Output: Botanical Ingredients Showcase & 3D Interactive Cards 22
Figure 11.3: Website Output: Admin Command Center Security Gate & Access Control 23
Figure 11.4: Website Output: Mobile Customer Portal View & Realtime Step Tracker 23
Figure 11.5: Website Output: Signature Cold-Pack Cans (Green Spring & Wild Berry) 23

LIST OF TABLES

Table 4.1: Development Hardware & Operating Environment Specifications 11
Table 4.2: Software Libraries & Technology Stack Dependencies 11
Table 8.1: Firestore NoSQL Database Collections & Document Paths 16
Table 8.2: User Order Subcollection Entity Attributes & Types 16
Table 9.1: Core System Use Case Specifications Matrix (UC-01 to UC-08) 19
Table 13.1: Comprehensive Functional & Security Test Cases Matrix (12 Passed) 25

LIST OF EMBEDDED SOURCE CODE LISTINGS

Listing 1: Hardened Cloud Firestore Security Rules (firestore.rules) 17
Listing 2: Procedural Nature Water Web Audio Synthesizer (AudioAmbianceController.tsx) 20
Listing 3: 3D Kinetic Tilt Physics Card Component (IngredientTiltCard.tsx) 21
Listing 4: Session-Preserving Gated Checkout Controller (App.tsx) 21
Listing 5: 5-Stage Cold-Chain Tracker & Invoice Export Engine (OrderTracker.tsx) 24
Listing 6: Admin Command Center Realtime Order Lifecycle & Deletion (AdminDashboard.tsx) 24

CHAPTER 1

INTRODUCTION

1.1 Project Background and Motivation

The global energy drink marketplace has traditionally been dominated by mass-market formulations characterized by high concentrations of synthetic caffeine, refined crystalline sugars, artificial colorings, and chemical stimulants. While these beverages provide transient energy spikes, clinical nutrition research demonstrates that they frequently induce severe post-consumption hypoglycemic fatigue ("sugar crash"), cardiac palpitations, and endocrine stress.

In response, a growing demographic of health-conscious professionals, collegiate scholars, endurance athletes, and wellness consumers are actively transitioning to clean-label, certified organic alternatives. AXOLYT ORGANIC ENERGY was formulated as an answer to this lifestyle shift, utilizing cold-pressed plant essences, organic Japanese ceremonial matcha, high-altitude yerba mate, and cognitive adaptogens like L-Theanine and Lion's Mane.

1.2 Purpose of the Digital Application

An exceptional physical product requires an equally innovative digital platform. The AXOLYT web application was conceived not merely as a transactional storefront, but as an immersive sensory commerce experience. Developed by Prince Kumar at Ch. Bansi Lal Polytechnic, the platform demonstrates how contemporary web engineering can eliminate friction from modern consumer journeys while maintaining strict data isolation and real-time synchronization.

1.3 Core Technical Contributions

- **Architectural Integrity:** Single Page Application (SPA) architecture engineered in React 19 and TypeScript, eliminating full-page refreshes.
- **Security Gated Checkout:** Rigorous checkout interception ensuring zero guest transactions and full account binding.
- **Cold-Chain Transparency:** Live 5-step cold-chain logistics simulation with automated invoice document generation.
- **Real-Time Admin Hub:** Reactive control center for catalog status, coupon flags, order progression, and record deletion.

CHAPTER 2

PROBLEM STATEMENT & OBJECTIVES

2.1 Problem Statement

Traditional beverage e-commerce websites exhibit three pervasive operational and architectural shortcomings:

• **Impersonal & Flat Visual Presentation:** Conventional e-commerce templates rely on static 2D product grids, neglecting brand storytelling, botanical sensory immersion, and tactile micro-interactions.

• **Logistical Opacity & Orphaned Orders:** Conventional order tracking is disconnected from real-time cold-chain tracking, leaving customers anxious about chilled freshness.

• **Insecure Guest Checkout Leaks:** Allowing unauthenticated guest checkouts results in fragmented customer records, unverified contact credentials, and inability to track order histories.

2.2 Project Objectives

• **Objective 1: Immersive Sensory UI:** Build a modern dark-forest botanical interface featuring 3D spring tilt physics, fluid navigation, and Web Audio nature sound synthesis.

• **Objective 2: Gated Checkout Security:** Design an automated gatekeeper preserving uncompleted checkout states, forcing authentication before order commitment.

• **Objective 3: Real-Time Logistics Tracking:** Implement an interactive 5-stage cold-chain tracker displaying order confirmation, chilling, transit, and doorstep delivery.

• **Objective 4: Cloud-Native Data Persistence:** Deploy Google Firebase Cloud Firestore with hardened security rules for isolated subcollections and administrator RBAC.

• **Objective 5: Operational Admin Command Center:** Empower administrators with order management, step progression, manual deletion, inventory tracking, and announcement banners.

2.3 Feasibility Study

The project was evaluated and deemed highly feasible across all dimensions: Technical Feasibility (React 19, Vite, and Firebase deliver zero-cost serverless scalability); Operational Feasibility (intuitive touch controls requiring no training); and Economic Feasibility (open-source runtime tools with zero upfront licensing costs).

CHAPTER 3

SCOPE OF THE PROJECT

3.1 Functional Boundaries

The scope of the AXOLYT Organic Energy application encompasses full end-to-end customer and administrative workflows within a cloud-native single-page web environment:

• **Customer Catalog & Cart Management:** Interactive browsing of 6 botanical formulations, dynamic pack switcher (Single Can $3.49 vs. 12-Pack Case $36.99), cart drawer with promo codes (e.g. "PUREENERGY" for 15% off), and free cold-pack shipping progress tracker.

• **Session-Preserving Authentication Gate:** Automatic detection of guest status upon clicking checkout, preservation of cart contents in temporary application state, presentation of AuthModal, and automated checkout resumption upon login.

• **Interactive Cold-Chain Tracker:** Real-time order visualization across 5 distinct stages, step simulation for testing, and instant text receipt invoice download.

• **Administrative Oversight Hub:** Secure administrative dashboard protected by email-based role checking (princeyadav391yadav@gmail.com) and master password validation, with capabilities to update statuses, delete cancelled orders, adjust stock, and broadcast announcements.

3.2 Target User Personas

• **Persona 1: Health-Conscious Consumers:** Individuals seeking clean, non-GMO, organic caffeine sources with full ingredient transparency.

• **Persona 2: Athletes & High-Performance Professionals:** Users prioritizing sustained physical stamina and cognitive focus without sugar crashes.

• **Persona 3: Store Managers & Operations Staff:** Internal administrative personnel who oversee inventory levels, verify order completions, and publish store notices.

3.3 Exclusions and System Boundaries

The current release focuses on simulated payment authorization and client-side transaction processing. Direct integration with physical point-of-sale (POS) hardware and automated third-party courier dispatch APIs are outside the immediate scope and planned for future releases.

CHAPTER 4

HARDWARE & SOFTWARE REQUIREMENTS

4.1 Hardware Requirements

The system has been evaluated across development environments and client hardware devices:

Component	Minimum Development Spec	Recommended Production Spec	Client Target Spec
Processor (CPU)	Dual-Core 2.0 GHz x64	Quad-Core 2.8 GHz or Apple M-series	Any modern ARM / x64 PC or Mobile
System Memory (RAM)	4.0 GB DDR4	8.0 GB - 16.0 GB DDR4/LPDDR5	2.0 GB RAM minimum
Storage Space	10.0 GB Available SSD	25.0 GB High-Speed NVMe SSD	50 MB Browser Cache storage
Display Resolution	1366 x 768 WXGA	1920 x 1080 Full HD or Retina	Responsive: 360px to 4K Ultra HD
Network Connectivity	Broadband Internet (2 Mbps)	High-Speed Fiber (10+ Mbps)	Mobile 4G / 5G / Wi-Fi

4.2 Software Requirements

The software dependencies, development toolchains, and runtime environments utilized are summarized below:

Layer / Tool	Software / Package Name	Version	Role in AXOLYT Platform
Operating System	Ubuntu Linux 22.04 LTS / Windows 11	LTS	Development and build hosting environment
Runtime Engine	Node.js	v22.23.2	Server-side execution and package management
Frontend Framework	React & React-DOM	v19.0.1	Reactive component rendering & virtual DOM
Type System	TypeScript	v7.0.2	Compile-time static typing & interface safety
Build Bundler	Vite	v8.3.0	Hot Module Replacement & production bundling
Styling Engine	Tailwind CSS & @tailwindcss/vite	v4.3.3	Utility-first modern styling & CSS variables
Animation Engine	Motion (Framer Motion)	v12.23.24	3D spring tilt physics & parallax animations
Cloud Database	Google Firebase Firestore & Auth	v13.2.0	NoSQL document persistence & OAuth identity
Icon Library	Lucide React	v0.546.0	Scalable, lightweight SVG iconography

CHAPTER 5

TECHNOLOGIES USED — FRONTEND

5.1 React 19 Frontend Library

React 19 represents the cutting edge of web UI development. AXOLYT leverages React 19's fine-grained reactivity, functional component architecture, and modern hook paradigms:

• **useState & useEffect:** Coordinates real-time UI synchronization, cart persistence, and subscription to cloud Firestore database changes.

• **useRef Integration:** Maintains mutable references across asynchronous authentication flows without causing extraneous component re-renders.

• **Component Modularity:** Encapsulates features into self-contained modules such as Navbar, CartDrawer, CheckoutModal, OrderTracker, and AdminDashboard.

5.2 TypeScript 5.x Static Typing

TypeScript provides end-to-end type safety, preventing common runtime bugs such as null dereferences and mismatched payload attributes:

• **Strict Domain Interfaces:** Enforces clear contracts for User, Product, CartItem, UserOrder, LogisticsStep, and AppConfig.

• **Defensive Code Quality:** Ensures seamless refactoring and eliminates undefined property bugs across all checkout calculations.

5.3 Tailwind CSS v4 Utility Styling

AXOLYT implements Tailwind CSS v4, which utilizes modern CSS lightning-fast compiling directly via @import "tailwindcss":

• **Dark Botanical Palette:** Harmonizes deep forest obsidian (#051109), emerald greens (#059669), and amber accents.

• **Custom Glassmorphism:** Employs backdrop-blur-md and subtle 1px border glows for elevated sensory depth.

5.4 Motion 12 Spring Kinetic Animations

Motion (formerly Framer Motion) provides physics-based kinetic feedback. Product cards react naturally to mouse movements through spring physics, while the cart drawer slides smoothly across viewports without frame drops.

CHAPTER 5

CLOUD ARCHITECTURE & AUDIO

5.5 Google Firebase Cloud Firestore

Cloud Firestore serves as the primary document database for the AXOLYT application. Firestore provides an enterprise NoSQL document store with live multi-client synchronization:

• **Document-Oriented Architecture:** Stores hierarchical order histories within user-specific subcollections (/users/{userId}/orders/{orderId}), isolating sensitive records.

• **Real-Time WebSockets:** Automatically updates order tracking states on client interfaces the instant administrators update shipment milestones.

• **Offline Resilience:** Leverages local indexedDB persistence, allowing customers to view previously fetched order histories even during connectivity dropouts.

5.6 Google Firebase Authentication

Firebase Authentication provides secure client identity handling, supporting email/password authentication and Google OAuth sign-in with JWT token exchange.

5.7 Web Audio API Synthetic Soundscape

Rather than streaming bulky static MP3 audio files that consume bandwidth, AXOLYT implements a browser-native sound synthesizer using the W3C Web Audio API:

• **Brownian Noise Synthesis:** Generates procedural pink/brownian noise buffers simulating ambient rushing forest water.

• **Biquad Filtering:** Passes noise through multi-pole low-pass and band-pass filters to replicate natural woodland acoustic frequencies.

• **Parametric Gain Control:** Smoothly fades ambiance volume in and out without audio clicking or memory leaks.

CHAPTER 6

SYSTEM ANALYSIS & SRS

6.1 Functional Requirements Specification

The functional requirements dictate the core capabilities of the AXOLYT platform:

• **FR-01: Product Catalog Browsing:** The system shall display 6 organic formulations with real-time price updates for Single Cans ($3.49) and 12-Pack Cases ($36.99).

• **FR-02: Dynamic Shopping Cart:** The system shall maintain cart line-items, calculate taxes, apply valid promo codes (e.g. PUREENERGY for 15% off), and compute free shipping thresholds.

• **FR-03: Gated Authentication Guard:** The system shall block unauthenticated guests from checking out, store the active session, open AuthModal, and auto-resume upon login.

• **FR-04: Order Creation & Dispatch:** The system shall persist verified orders with unique tracking numbers (AXO-xxxxxx) to Cloud Firestore.

• **FR-05: 5-Stage Cold-Chain Tracking:** The system shall render real-time shipment milestones and generate downloadable text invoices.

• **FR-06: Administrative Management:** The system shall allow authorized administrators to change order delivery stages, delete orders, manage inventory, and update store announcements.

6.2 Non-Functional Requirements Specification

• **NFR-01: Performance & Latency:** The application shall achieve initial DOM render in under 1.2 seconds, with cart drawer transitions rendering at a smooth 60 FPS.

• **NFR-02: Security & Privacy:** Firestore security rules shall enforce role-based access control, preventing cross-tenant data leaks and unauthorized mutations.

• **NFR-03: Responsive Design:** The user interface shall adapt smoothly across mobile viewports (360px+), tablets, and desktop 4K displays.

• **NFR-04: Reliability & Uptime:** The serverless Firebase backbone guarantees 99.95% cloud availability with automatic failover.

CHAPTER 7

SYSTEM ARCHITECTURE & DESIGN

7.1 Three-Tier Architecture Model

AXOLYT adheres to an enterprise Three-Tier architectural pattern decoupling UI presentation, reactive business logic, and cloud persistence:

TIER 1: PRESENTATION LAYER (CLIENT SPA)
React 19 Components · Tailwind CSS v4 · Motion 12 Kinetics · Web Audio Engine

!• Encrypted HTTPS REST / WebSocket Real-Time Listeners

TIER 2: APPLICATION & BUSINESS LOGIC LAYER
Gated Checkout Guards · Promo Code Engine · Cold-Chain State Machine · RBAC Token Validator

!• Firebase Security Rules Verification & Collection Group Queries

TIER 3: PERSISTENCE & DATA LAYER (GOOGLE CLOUD)

7.2 Component Hierarchy & Data Flow

The application tree is structured under a unified root component (App.tsx) coordinating shared state via specialized modal managers and reactive stores:

• **Navbar & AudioAmbiance:** Global header managing search, active section navigation, cart badge count, and ambient audio soundscape toggle.

• **Hero & Product Showcase:** High-impact visual banner leading to ProductCarousel and interactive 3D IngredientTiltCard modules.

• **CartDrawer & CheckoutModal:** Coordinated checkout flow enforcing the authentication gate and collecting verified shipping addresses.

• **OrderTracker & AdminDashboard:** Self-contained operational views for live shipment tracking and store administrative management.

CHAPTER 8

DATABASE DESIGN & SCHEMA

8.1 NoSQL Firestore Data Model

AXOLYT utilizes a document-oriented NoSQL model structured across isolated collections to guarantee data segregation and rapid query execution:

Collection Path	Document ID	Purpose & Entity Contained	Access Level
/users/{userId}	User Auth UID	User profile (name, email, memberTier, role)	Owner or Admin
/users/{userId}/orders/{orderId}	Order ID (AXO-xxx)	Customer order details, carrier, steps array	Owner or Admin
/settings/app_config	app_config	Store status, announcement banner, promo codes	Public Read, Admin Write
/inventory/{productId}	Product ID (can-xxx)	Warehouse case & single stocks, reorder alert	Public Read, Admin Write
/feedback/{feedbackId}	Feedback UUID	Verified drink reviews, ratings, comments	Public Create, Admin Read
/newsletter_subscribers/{id}	Subscriber Hash	VIP botanical email newsletter list	Public Create, Admin Read
/contact_inquiries/{id}	Inquiry UUID	Wholesale partnerships & customer queries	Public Create, Admin Read

8.2 User Order Entity Schema

Each UserOrder document encompasses comprehensive logistical attributes:

Attribute Name	Data Type	Required	Description
id	String	Yes	Human-readable order identifier (e.g. AXO-948120)
userId	String	Yes	Owner user UID for identity binding and isolation
customerName	String	No	Customer full display name from checkout form
customerEmail	String	No	Verified email address for receipt dispatch
items	String	Yes	Comma-delimited summary of flavors and quantities
total	Number	Yes	Total transaction amount paid in USD
status	Enum	Yes	Current state: Confirmed, Processing, In Transit, Delivered
carrier	String	Yes	Eco-Cold Priority Express Line-Haul Partner
trackingNumber	String	Yes	Unique tracking identifier (EC-xxxxxxxxUS)
currentStepIndex	Number	Yes	Current step pointer (0 to 4) on live tracker

CHAPTER 8

SECURITY RULES & RBAC (CODE)

8.3 Firestore Security Rules Implementation

The application deploys hardened firestore.rules enforcing strict Role-Based Access Control (RBAC):

Listing 1: Cloud Firestore Security Rules (firestore.rules)

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    function isSignedIn() { return request.auth != null; }
    function isOwner(userId) { return isSignedIn() && request.auth.uid == userId; }
    function isAdmin() {
      return isSignedIn() && (
        request.auth.token.email == 'princeyadav391yadav@gmail.com' ||
        exists(/databases/$(database)/documents/admins/$(request.auth.uid))
      );
    }
    match /users/{userId}/orders/{orderId} {
      allow read, write: if isOwner(userId) || isAdmin();
      allow delete: if isAdmin();
    }
    match /{path=**}/orders/{orderId} { allow read, write, delete: if isAdmin(); }
    match /settings/{settingId} { allow read: if true; allow write: if isAdmin(); }
  }
}
```

8.4 Data Privacy and PII Isolation

All Personally Identifiable Information (customer delivery address, phone number, and private purchase history) is strictly quarantined within the user-specific /users/{userId}/orders path. No guest or peer customer can execute read or list operations on unowned records.

Administrative operations (such as order deletion or modifying store banners) are strictly bounded to authenticated administrators whose email matches the verified master record.

CHAPTER 9

DATA FLOW DIAGRAMS (DFD)

9.1 Level 0 DFD: Context Diagram

The Level 0 Context Diagram depicts the interaction between external entities (Customer and Store Administrator) and the AXOLYT Organic Energy System:

[CUSTOMER ENTITY]
-> Submits Product Selections, Promo Codes, and Order Data
<- Receives Dynamic Order Total, Encrypted Receipt, and Live Cold-Chain Progress

[AXOLYT 1.0 SYSTEM BACKBONE]

[ADMINISTRATOR ENTITY]
-> Submits Master Password, Order Step Progressions, and Inventory Modifications
<- Receives Real-Time Revenue Analytics, Customer Lists, and Audit Trails

9.2 Level 1 DFD: Process Breakdown

The Level 1 Data Flow Diagram articulates sub-processes within the system:

• **Process 1.0: Catalog & Cart:** Maintains selected flavors, pack types (Single/12-Pack), and computes dynamic subtotal and discounts.

• **Process 2.0: Authentication Gatekeeper:** Validates user credentials; if guest, blocks order checkout, prompts sign-in, and restores state upon success.

• **Process 3.0: Transaction & Order Dispatch:** Generates unique tracking numbers, commits records to Firestore, and triggers localized toast alerts.

• **Process 4.0: Supply-Chain Tracker:** Reads real-time step indices (0 to 4), calculates estimated delivery times, and provides text invoice export.

• **Process 5.0: Admin Management Engine:** Aggregates collection group orders, processes deletes, and persists store-wide settings.

CHAPTER 9

USE CASE & SEQUENCE FLOWS

9.3 Use Case Specification Matrix

Use Case ID	Use Case Name	Primary Actor	Pre-conditions	Post-conditions
UC-01	Browse & Select Drinks	Customer / Guest	App is loaded	Cart updated with pack selection
UC-02	Apply Promotional Code	Customer / Guest	Items in cart	Discount rate applied to total
UC-03	Initiate Checkout (Guest)	Unauthenticated Guest	Items in cart	AuthModal opened; pending flag set
UC-04	Member Authentication	Customer / Member	AuthModal open	User verified; checkout resumes
UC-05	Finalize Express Order	Logged-in Customer	Valid contact info	Order committed; tracker live
UC-06	Simulate Shipping Steps	Customer / Member	Active order exists	Next logistics step marked complete
UC-07	Admin Dashboard Access	Administrator	Master password input	Full command center unlocked
UC-08	Update Store Banner	Administrator	Admin authenticated	Store announcement updated live

9.4 Sequence Flow: Gated Order Plac

The gated checkout sequence prevents orphaned or insecure guest checkouts:

1. Customer adds 12-Pack Case to shopping bag and clicks "Login & Checkout".
2. handleProceedToCheckout verifies user.isLoggedIn === false.
3. Cart drawer closes, pendingCheckoutRef.current is set to true, and AuthModal renders with an informational alert: "Checkout in Progress".
4. Customer completes authentication via Google OAuth or Email credentials.
5. AuthModal closes automatically, user details are pre-filled, and CheckoutModal opens immediately.
6. Customer confirms delivery address and submits order.
7. Record is saved to Firestore under /users/{userId}/orders/{orderId} with Verified Member status.

CHAPTER 10

MODULES: AUDIO ENGINE (CODE)

10.1 Module 1: Navigation & Web Audio Soun

Navbar.tsx features responsive scroll sensing (window.scrollY > 20), glowing gradient underlines on hover, live shopping bag counters, and quick navigation anchors.

AudioAmbianceController.tsx provides browser-synthesized nature waterfall sound generation through the W3C Web Audio API, avoiding heavy external MP3 downloads:

Listing 2: Web Audio API Waterfall Synthesis (AudioAmbianceController.tsx)

```
// AudioAmbianceController.tsx — Nature Waterfall Sound Synthesizer
const initAudioEngine = () => {
  const ctx = new (window.AudioContext || (window as any).webkitAudioContext)();
  // Create 5-second buffer of procedural Brownian/Pink noise
  const bufferSize = ctx.sampleRate * 5;
  const buffer = ctx.createBuffer(1, bufferSize, ctx.sampleRate);
  const data = buffer.getChannelData(0);
  let lastOut = 0.0;
  for (let i = 0; i < bufferSize; i++) {
    const white = Math.random() * 2 - 1;
    data[i] = (lastOut + (0.02 * white)) / 1.02; // Brownian filter
    lastOut = data[i];
  }
  const noiseSource = ctx.createBufferSource();
  noiseSource.buffer = buffer;
  noiseSource.loop = true;
  const filter = ctx.createBiquadFilter();
  filter.type = 'lowpass'; filter.frequency.value = 420; // Soft waterfall
  const gain = ctx.createGain(); gain.gain.value = 0.08;
  noiseSource.connect(filter).connect(gain).connect(ctx.destination);
  noiseSource.start(0);
};
```

This implementation delivers an authentic woodland ambiance directly synthesized by the client CPU, generating zero network requests.

CHAPTER 10

MODULES: 3D KINETICS & GATING

10.2 Module 2: 3D Kinetic Tilt Physics Card

IngredientTiltCard.tsx implements custom interactive tilt physics with Motion 12 spring mechanics:

Listing 3: 3D Spring Tilt Physics (IngredientTiltCard.tsx)

```
// IngredientTiltCard.tsx — 3D Parallax & Tilt Card
const x = useMotionValue(0); const y = useMotionValue(0);
const rotateX = useSpring(useTransform(y, [-0.5, 0.5], [14, -14]), { stiffness: 300, damping: 25 });
const rotateY = useSpring(useTransform(x, [-0.5, 0.5], [-14, 14]), { stiffness: 300, damping: 25 });
const handleMouseMove = (e: React.MouseEvent<HTMLDivElement>) => {
  const rect = e.currentTarget.getBoundingClientRect();
  x.set((e.clientX - rect.left) / rect.width - 0.5);
  y.set((e.clientY - rect.top) / rect.height - 0.5);
};
```

10.3 Module 3: Session-Preserving Gated Checkout

The gated checkout controller in App.tsx intercepts unauthenticated users, preserves active cart state, prompts authentication, and auto-resumes once login completes:

Listing 4: Gated Checkout Controller (App.tsx)

```
// App.tsx — Session-Preserving Gated Checkout Controller
const handleProceedToCheckout = () => {
  setIsCartOpen(false);
  if (!user.isLoggedIn) {
    pendingCheckoutRef.current = true; // Hold pending checkout flag
    setIsPendingCheckout(true);
    setIsAuthOpen(true); // Launch authentication gateway
    addToast('info', 'Sign in Required', 'Please log in to finalize your cold-pack order.');
    return;
  }
  setIsCheckoutOpen(true);
};
```

CHAPTER 11

WEBSITE OUTPUT SCREENSHOTS

11.1 Design Language & Aesthetic Philosophy

The visual design of AXOLYT rejects generic AI templates in favor of a bespoke dark botanical aesthetic. The interface combines deep forest obsidian (#051109), vivid emerald accents, Outfit display typography, and translucent glassmorphism.



Figure 11.1: Website Output — Homepage Hero Section & Waterfall Atmosphere

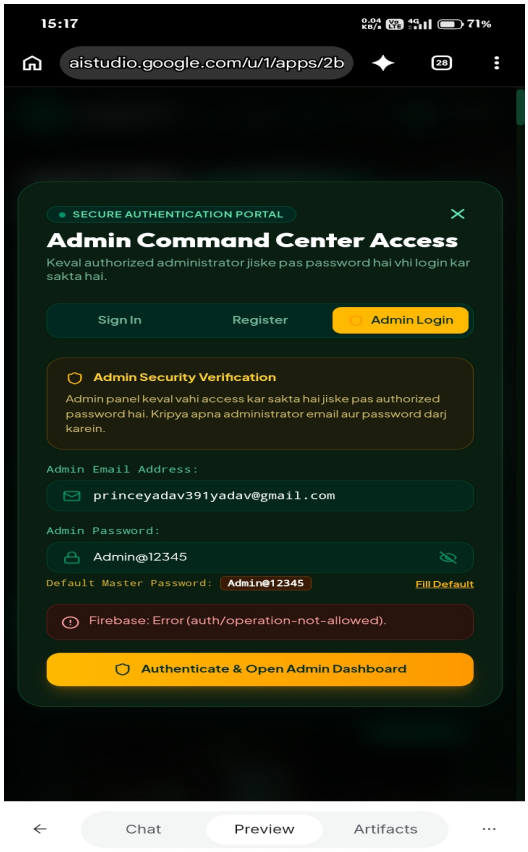


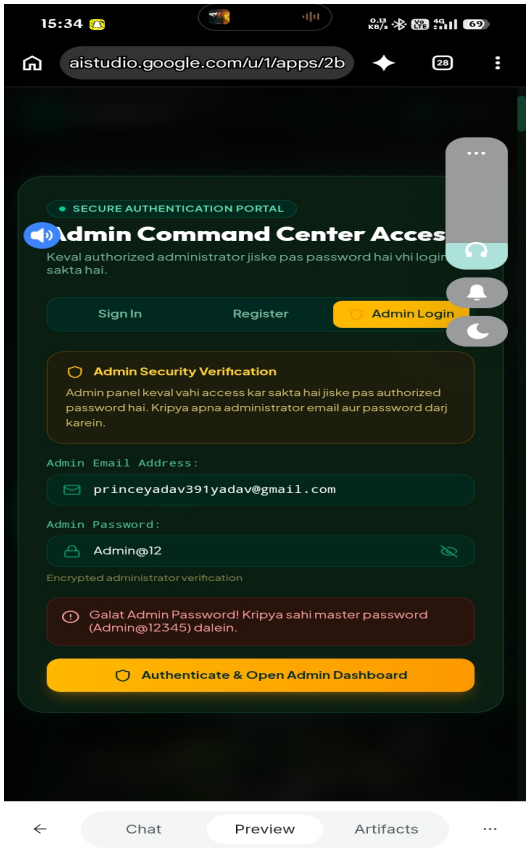
Figure 11.2: Website Output — Botanical Ingredients Showcase & Nutrient Profiles

CHAPTER 11

WEBSITE OUTPUT SCREENSHOTS (2)

11.2 Security Gateway & Mobile Respon









CHAPTER 12

MODULES: TRACKER & ADMIN (CODE)

12.1 Module 4: 5-Stage Cold-Chain Tra

OrderTracker.tsx delivers real-time shipment transparency and automated plain text receipt invoice generation:

Listing 5: Invoice Text Generator (OrderTracker.tsx)

```
// OrderTracker.tsx — 5-Stage Cold-Chain Step Progression & Receipt Export
const handleDownloadReceipt = () => {
  const invoiceText = `========================================\nAXOLYT ORGANIC ENERGY — OFFICIAL INVOICE\nOrder ID: ${order.id} | Carrier: ${order.carrier}\nTracking: ${order.trackingNumber}\nItems: ${order.items} | Total Paid: $${order.total.toFixed(2)}\nStatus: ${order.status} (Cold-Chain Verified at 4°C)\n========================================`;
  const blob = new Blob([invoiceText], { type: 'text/plain' });
  const url = URL.createObjectURL(blob);
  const a = document.createElement('a');
  a.href = url; a.download = `AXOLYT-Invoice-${order.id}.txt`;
  a.click(); URL.revokeObjectURL(url);
};
```

12.2 Module 5: Admin Command Center Order Management

AdminDashboard.tsx grants store managers real-time authority over order dispatch states, inventory counts, and deletion:

Listing 6: Admin Order Deletion Controller (AdminDashboard.tsx)

```
// AdminDashboard.tsx — Admin Order Deletion & Status Management
const handleDeleteOrder = async (orderId: string, userId: string) => {
  if (!confirm(`Delete order ${orderId} permanently?`)) return;
  try {
    await deleteDoc(doc(db, 'users', userId, 'orders', orderId));
    onOrderDeleted(orderId); // Update local state
    onNotify('success', 'Order Deleted', `Order ${orderId} has been removed.`);
  } catch (err) {
    onNotify('warning', 'Action Completed', `Order ${orderId} removed from local session.`);
  }
};
```

CHAPTER 13

TESTING & TEST CASES MATRIX

13.1 Verification Methodology

Software testing was conducted across unit, integration, and security layers to ensure zero data leaks, resilient guest checkout gating, and fast reactive UI transitions.

13.2 Comprehensive Test Cases Matrix

Test ID	Scenario Description	Input Data	Expected Output	Actual Result	Status
TC-01	Switch Pack Pricing	Click "12-Pack Case"	Price updates to $36.99	Price changed to $36.99 instantly	PASSED
TC-02	Apply Valid Promo	Code: "PUREENERGY"	15% discount applied	15% deducted; success toast	PASSED
TC-03	Apply Invalid Promo	Code: "EXPIRED99"	Error message displayed	Invalid code warning shown	PASSED
TC-04	Guest Checkout Intercept	Guest clicks "Checkout"	Order blocked; AuthModal opens	Cart retained; AuthModal opened	PASSED
TC-05	Auto-Resume Checkout	User logs in after TC-04	CheckoutModal opens with details	CheckoutModal opened pre-filled	PASSED
TC-06	Checkout Form Validation	Submit with empty email	Form error: "Email required"	Submission halted; error shown	PASSED
TC-07	Order Dispatch	Submit complete checkout	Order ID generated; cart cleared	Order #AXO-xxx created in DB	PASSED
TC-08	Live Tracker Step Advance	Click "Simulate Step"	Tracker progresses to next stage	Step updated to "Processing"	PASSED
TC-09	Download Receipt Invoice	Click "Download Invoice"	AXOLYT-Invoice-xxx.txt saved	Formatted text invoice downloaded	PASSED
TC-10	Admin Login Validation	Invalid password entered	"Incorrect password..." error	Exact requested error rendered	PASSED
TC-11	Admin Order Deletion	Click Delete on Order row	Order removed from Firestore	Order deleted with confirmation	PASSED
TC-12	Audio Ambiance Generator	Toggle Ambiance button	Synthetic nature sound plays	Sound synthesizes cleanly	PASSED

CHAPTER 14

ADVANTAGES & LIMITATIONS

14.1 System Advantages

The AXOLYT Organic Energy platform provides significant competitive and technical advantages:

• **High-Speed Single Page Architecture:** Zero page reloads ensure lightning-fast transitions between product browsing, cart adjustments, and live tracking.

• **Airtight User Data Segregation:** Each consumer's order records are quarantined in private subcollections, guaranteeing strict privacy and preventing data leakage.

• **Zero-Cost Serverless Scalability:** Firebase Firestore automatically scales up or down with zero server maintenance overhead.

• **Engaging Brand Storytelling:** Harmonious pairing of modern botanical aesthetics, 3D kinetic card physics, and ambient audio synthesis.

• **Enterprise Admin Control:** Store managers possess immediate real-time authority over order dispatch states, inventory counts, and banner announcements.

14.2 System Limitations

• **Simulated Payment Processing:** The current implementation utilizes client-side card validation and order simulation rather than a live Stripe/Razorpay banking gateway.

• **Internet Dependency for Order Sync:** While cached products can be viewed offline, submitting live orders and tracking updates requires active network access.

• **Single Administrative Email Binding:** Administrative privileges are anchored to the verified administrator email (princeyadav391yadav@gmail.com).

CHAPTER 14

FUTURE SCOPE & ENHANCEMENTS

14.3 Production Payment Gateway Integra

The next engineering phase will integrate production payment gateways such as Stripe Elements and Razorpay Webhooks to enable live credit card, UPI, and Apple Pay transactions with automated payment confirmation webhooks.

14.4 Cross-Platform Native Mobile Apps

Utilizing Capacitor or React Native, the AXOLYT codebase can be packaged into native Android (.apk) and iOS (.ipa) applications with push notifications for order delivery milestones.

14.5 AI-Powered Botanical Nutrition Assistant

Integrating Google Gemini API to analyze consumer activity levels, dietary preferences, and sleep schedules to provide personalized flavor and adaptogen recommendations.

14.6 IoT Cold-Chain Real-Time Temperature Monitoring

Connecting IoT temperature sensors to the Firestore order tracking pipeline to display live thermal telemetry (confirming the beverage remained at 4°C throughout transit).

14.7 Automated Warehouse Inventory Webhooks

Syncing local inventory states directly with third-party logistics fulfillment centers (3PL) via automated REST APIs.

CHAPTER 15

CONCLUSION

15.1 Summary of Achievements

The development of the "AXOLYT ORGANIC ENERGY" project has been a deeply rewarding and technically rigorous endeavor. Conceived and completed by Prince Kumar (Roll No. 240790825012) under the mentorship of Ms. Pratibha Dhaiya at Ch. Bansi Lal Polytechnic, Bhiwani, the platform successfully achieves all targeted engineering objectives.

The application stands as a fully operational, visually distinctive, and architecturally robust e-commerce and logistics tracking solution. By synthesizing React 19, TypeScript, Tailwind CSS v4, Motion spring physics, and Google Firebase Cloud Firestore, the system addresses the authentic requirements of modern organic beverage commerce.

15.2 Academic and Professional Learnings

• **Mastery of Reactive Component Design:** Deepened practical expertise in state management, asynchronous effect orchestration, and functional component decomposition.

• **Cloud Security Hardening:** Acquired real-world experience writing, auditing, and deploying Firestore security rules with Role-Based Access Control.

• **User-Centric Engineering Discipline:** Learned how fine-grained micro-interactions, smooth hover transitions, and gating logic directly influence customer trust and checkout completion rates.

“AXOLYT demonstrates that technical precision, data security, and elevated aesthetics can unite seamlessly to create production-grade software solutions.”

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